



Mechanisms of antifungal resistance and developments in alternative strategies to combat *Candida albicans* infection

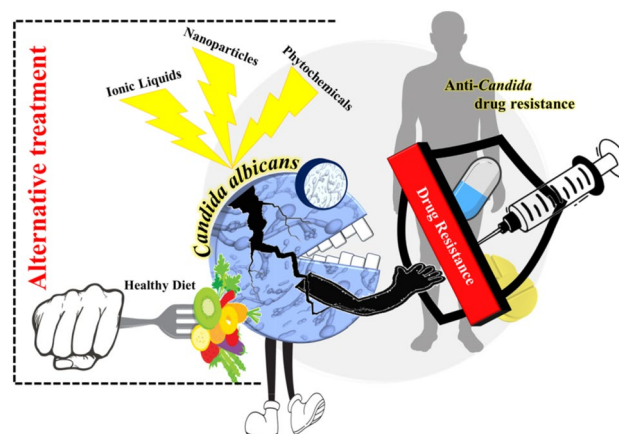
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Received: 20 November 2023 / Revised: 21 December 2023 / Accepted: 29 December 2023 / Published online: 13 February 2024
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Abstract

Candida albicans is a commensal fungus that infects the humans and becomes an opportunistic pathogen particularly in immuno-compromised patients. Among the *Candida* genus, yeast *C. albicans* is the most frequently incriminated species and is responsible for nearly 50–90% of human candidiasis, with vulvovaginal candidiasis alone, affecting about 75% of the women worldwide. One of the significant virulence traits in *C. albicans* is its tendency to alternate between the yeast and hyphae morphotypes, accounting for the development of multi-drug resistance in them. Thus, a thorough comprehension of the decision points and genes controlling this transition is necessary, to understand the pathogenicity of this, naturally occurring, pernicious fungus. Additionally, the formation of *C. albicans* biofilm is yet another pathogenesis trait and a paramount cause of invasive candidiasis. Since 1980 and in 90 s, wide spread use of immune-suppressing therapies and over prescription of fluconazole, a drug used to treat chronic fungal infections, triggered the emergence of novel anti-fungal drug development. Thus, this review thoroughly elucidates the diseases associated with *C. albicans* infection as well as the anti-fungal resistance mechanism associated with them and identifies the emerging therapeutic agents, along with a rigorous discussion regarding the future strategies that can possibly be adopted for the cure of this deleterious pathogen.

Graphical Abstract



Keywords *Candida albicans* · *Candida* infections · Mechanism of pathogenesis · Anti-fungal resistance · Alternative anti-*Candida* agents

Communicated by Yusuf Akhter.

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Introduction

The occurrence of fungal infections, especially candidemia (*Candida* blood stream infection) has aggravated significantly, leading to an increased number of casualties in

the developed countries. Invasive candidiasis has made its appearance as a prime mover of morbidities and mortalities associated with the health-care environment (Costa-de-Oliveira and Rodrigues 2020; McCarty and Pappas 2016). Various *Candida* spp. are responsible for human pathogenicity, however, most of the invasive infections are brought about by 5 major species of *Candida*, namely, *C. albicans*, *C. glabrata*, *C. tropicalis*, *C. parapsilosis* and *C. krusei* (McCarty and Pappas 2016; Bhattacharya et al. 2020; Fisher et al. 2022a; Czechowicz et al. 2022). Moreover, *C. auris*, a formerly intermittent organism, has recently come into light as a major pathogen in certain parts of the globe (Chowdhary et al. 2017). The colonization of these *Candida* spp. is imperative for subsequent progression of the infection. Generally, all *Candida* spp. are commensal yeasts, making a part of healthy human skin as well as gut microbiota and thus are apprehensible in up to 60% of normal individuals (McCarty and Pappas 2016; Fisher et al. 2022a). Invasive candidiasis is considered to be a disorder with multitudinous clinical expressions affecting any organ in the human body, since, each *Candida* spp. bears its own unique invasive potential, virulence characteristics, as well as susceptibility to anti-fungal agents. However, *C. albicans* is reported to be the most accepted pathogen in several clinical settings with the other *Candida* spp. collectively representing more than 50% of the isolates from the human bloodstream in many regions (McCarty and Pappas 2016; Bhattacharya et al. 2020; Fisher et al. 2022a; Czechowicz et al. 2022).

C. albicans is considered to be an opportunistic fungal pathogen occurring as a common member of the normal microflora in the human digestive tract. This species is just one of the 200 species in the genus *Candida* accounting up to 75% of all *Candida* infections. *C. albicans* is known to be one of the foremost eukaryotic pathogens having its genome sequenced with its clinical isolates comprising of a 16 Mb genome, with 33.3% Guanine-cytosine (GC) content, 132 non-coding RNAs, a large number of transposons, 8 diploid chromosomes and 6735 (haploid) open reading frames (ORFs). Its genome is considered to be extremely plastic with a great extent of heterozygosity, intra-chromosome recombination and aneuploidy, enabling stress adaptation and affecting the drug sensitivity by influencing the copy number of key drug resistance alleles. *C. albicans* are known to exhibit a parasexual cycle giving rise to tetraploid zygotes through the fusion of two diploids of opposite mating types under stressful conditions. Such mating behaviour is advantageous in evolving resistance against antifungal compounds (Gow and Yadav 2017). Usually, the innate and acquired host defense mechanisms act harmoniously over the resident bacterial flora fostering the commensal growth of *Candida* organisms which might turn up, to being pathogenic on even a slight modification of the host defense system or the ecological environment of the host. This class of

fungi may colonize in a healthy individual at an early age either through the birth canal, or during nursing or from food and the common sites of asymptomatic colonization in the human body involves the oral cavity, rectum and vagina. A prolonged invasion in the human body is accountable for evoking the circulating immunoglobulin G (IgG) and mucosal secretory immunoglobulin A (S-IgA) antibodies to restrict the superficial growth of the fungus at the mucosal sites (Rabaan et al. 2023). *C. albicans* is known to be dimorphic in nature and multiplies as oval-shaped budding yeast, commonly referred to as a blastophore. Other phenotypes may include the pseudo-hyphae or true hyphae forms, as important morphotypes, forming a part of the infected tissue. The transition of *Candida* from yeast to hyphae form is a crucial factor associated with its virulence along with several other factors like adhesion that promotes the attachment of the fungal cells to the host cells, hydrolytic enzymes (e.g., proteinases) that increase adhesion property promoting tissue devastation (Lopes and Lionakis 2022) and molecules responsible for modulating the immune cell functions (Bohner et al. 2022).

The crucial challenges related to the management of the spread of the *C. albicans* infection involve its prevention, early detection and swift treatment with suitable systemic anti-fungal remedies. Routine fungal cultures are the sole diagnostic tools for the rapid establishment of the disease due to the dearth of other rapid diagnostic assays. The limitations of such detection tools being low sensitivity and empirical evidence, lead to unnecessary prescription of anti-fungals even in individuals not suffering from the invasive infection. This further detains the treatment with appropriate anti-fungal therapy for the infected patients. Thus, a delayed diagnosis and treatment could result in unexpectedly worse clinical outcomes.

There are enthralling evidences of association of *Candida* with other oral bacterial species, encouraging adhesion followed by colonization via mixed-species communities (Bohner et al. 2022). Hence, circumventing the proliferation of such fungal ailments is a pressing priority. In this regard, the study of novel anti-fungal drugs lags behind due to lack of seriousness towards the fungal pathogens. Up to 30 long years, amphotericin B was the only source of treatment associated with the fungal infections followed by the discovery of imidazole and triazoles out of which fluconazole gained popularity as sole anti-fungal drug during 1980s (Cavassin et al. 2021). The over-prescription of these drugs resulted in the emergence of drug-resistant pathogenic strains, challenging the existing anti-fungal regimen.

Thus, this review considers the etiology and pathology of several disease conditions arising due to *C. albicans* infection describing the clinical manifestations, adhesion and biofilm formation as a consequence of drug resistance. Additionally, this review describes various alternatives of the existing anti-fungal treatment by considering

the comprehension of the studies of microbial colonization mechanism, thereby assisting the development of effective methodology in prevention and control of *Candida* infections.

Pathogenicity of *Candida* infections and the transcriptomic regulation of the biofilm formation

The systemic infection of *C. albicans* thrives in a sluggish manner with no visible signs and symptoms. The developed cases are marked by sensation of flu, constipation, cough, fever, chest pain, chills weight loss and issues with breathing. As a consequence of yeast over growth, the immune system mechanism is depleted, damaging the normal intestinal lining, leading to elevated absorption of yeast cells, wide range of toxins and other substances, resulting in the build out of “Yeast Syndrome”. Women patients are more prone to yeast infections by eight folds than men, owing to increased

levels of estrogen and intake of birth control pills, with vaginal yeast infection, accompanied by bladder infections being a major cause of concern worldwide (Martins et al. 2014).

Customarily, various molecular and biochemical diagnostic techniques including, blood analysis as well as urine testing are employed to measure the severity of the yeast infections. Lately, several advancements made in the diagnostic platforms have made its diagnosis quite swift and facile, using techniques such as fluorescence in situ hybridization (PNA-FISH test) that distinguishes between several *Candida species*, rapidly (Oliveira Santos et al. 2018). Additionally, immunological and genetic techniques such as Enzyme-linked immunosorbent assay (ELISA) and Polymerase Chain Reaction (PCR) are used to diagnose invasive candidiasis and to differentiate between *C. albicans* and *C. dubliniensis* in circumstances where ever differentiation is required.

The transition from commensalism to pathogenesis, in *C. albicans* occurs in three stages, within a period of 24–48 h, namely, adhesion, invasion and damage. This transition is fostered by the up-regulation of the genes *ALS3*, *HWP1*, *ECE1*, *SOD5*, *PHR1*, *PRA1*, and *RBT1*, respectively. The

Table 1 Commercially available anti-fungal agents with their possible mechanism of action and cause of resistance

Class of drug	Commercially available drugs	Cause of Resistance	References
Azoles	Imidazole	Drug efflux is carried out as a result of reduced affinity in Erg11 protein caused by mutations, as well as upregulation of multi-drug transporter genes and a change in a particular stage in the ergosterol bio-synthetic pathway. The cause of fluconazole resistance is the over-expression of <i>cdr1</i> , <i>cdr2</i> , <i>mdr1</i> genes that overstimulate the ABC transporters pump and modification of the enzyme encoded by the <i>erg11</i> gene	Pemán et al. 2011)
	Miconazole		
	Oxiconazole		
	Econazole		
	Ketoconazole		
	Tioconazole		
	Clotrimazole		
	Triazole		
	FLC		
	Posaconazole		
	Itraconazole		
	Terconazole		
	voriconazole		
Polyenes	Amphotericin B (AmpB)	Resistance to this class of drugs may be a consequence of altered membrane lipids, particularly sterols. Transformation of the phospholipids along with augmented catalase activity and reduced susceptibility to oxidative damage may also contribute in drug resistance	Pristov and Ghannoum 2019)
	Natamycin		
	Nystatin		
Pyrimidine Analogs	5-fluorocytosine (5-FC)	(i) Changes in purine-cytosine permease (FCY2) gene, which is responsible for the uptake of the drug into the cell (ii) Changes in the structure of the enzyme cytosine deaminase (FCY1) gene, responsible for the conversion to 5-fluorouracil (iii) Changes in uracil phosphoribosyl transferase, responsible for the transformation of 5-fluorouracil to 5-fluorouridine monophosphate (gene FUR1)	Fisher et al. 2022b)
	5-fluorouracil (5-FU)		
	Griseofulvin		
Allylamine, Thiocarbamate and Morpholine	Fenpropimorph Amorolfine Terbinafine	Alteration in gene ERG1 is the major cause of Allylamine resistance	Kathiravan et al. 2013)
Echinocandins	Caspofungin Micafungin Anidulafungin	Changes in echinocandin affinity for 1,3 glucan synthetase causes caspofungin resistance	Perlin 2015)

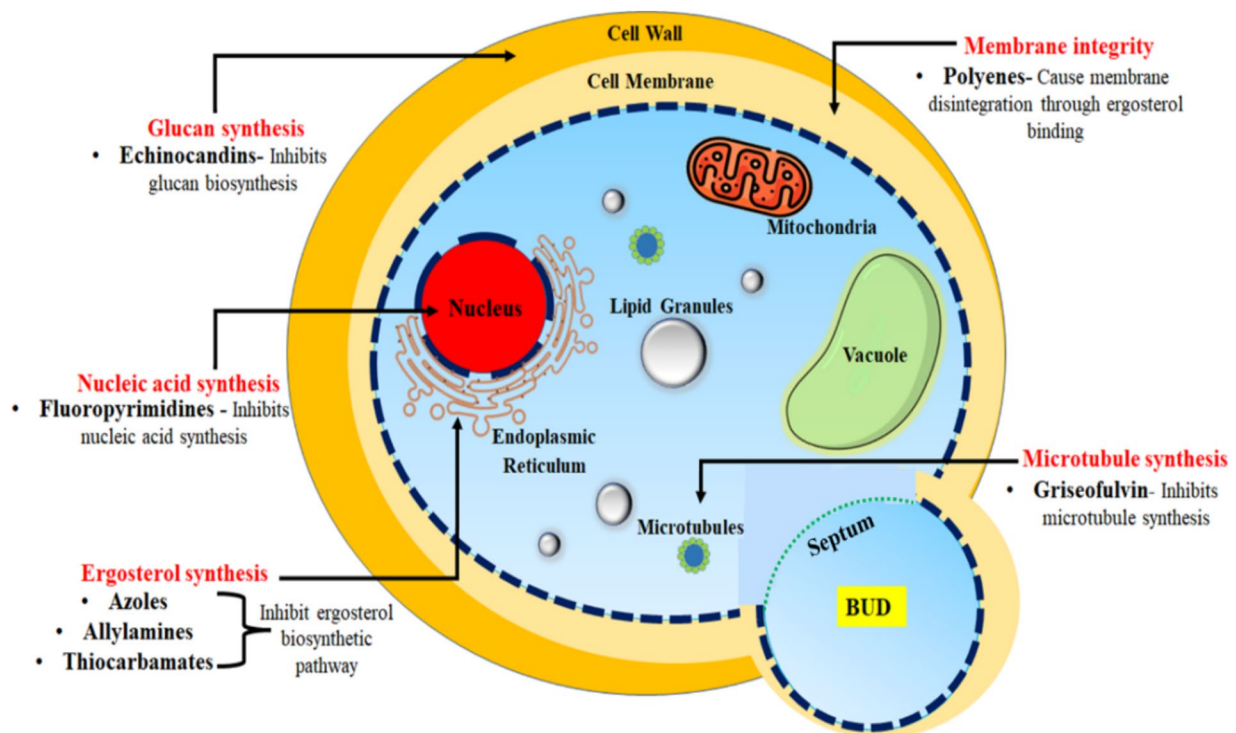


Fig. 1 Point of action of various antifungal agents

hyphae formation, promoted by the release of Hwp1 and Als3 proteins, encourages the adhesion of the fungal cells to the host or substrate surfaces, damaging the host cells, which is one of the crucial characteristics of pathogenesis associated with *Candida* infections. Two major proteins, namely, aspartyl proteinases (Saps) and phospholipases are key participants in bringing out the hyphal invasion within the host cells. The penetration of these filamentous structures deep through the epithelial layer of the host cells causes the destruction of the host tissues (Chin et al. 2016; Chow et al. 2021).

More than 1,000 genes are accountable for governing the formation of biofilm in *C. albicans*, however, the normal phase of biofilm development constitutes of majorly six master transcription regulators, such as, *Efg1*, *Bcr1*, *Brg1*, *Ndt80*, *Tec1*, and *Rob1* (Chong et al. 2018), the roles of which are elaborated in the table. The principle transcription regulators involved in *C. albicans* biofilm formation are *Efg1*, *Bcr1*, *Brg1*, *Ndt80*, *Tec1* and *Rob1*, respectively. Furthermore, *Efg1* and *Tec1* are also responsible for regulating the cell morphology (Panariello et al. 2017), while, *Ndt80*, is the major contributor in biofilm formation. The functions associated with *Bcr1* are yet to be delineated thoroughly. Additionally, *Rfx2*, *Gal4* and *Flo8* are recently established to play distinctive roles in the propagation of biofilm overtime. Finkel et al. (Finkel et al. 2012) were the first to discover the existence of 29 transcription factors regulating *C. albicans*

adhesion to a silicone base. Among these, only *Ace2p*, *Arg81p*, *Bcr1p*, and *Snf5p* mutants demonstrated anti-adherence properties under in vitro conditions (Pristov and Ghanoun 2019). Additionally, the deletion of *Zcf28p*, *Zfu2p*, and *Crz2p* impedes biofilm formation on catheters. Furthermore, five crucial adherence regulators, namely, *ARC18*, *PMT1*, *MNN9*, *SPT7* and *orf19.831* were delineated, transcriptional suppression of which tends to inhibit the adherence of *C. albicans*, with *ARC18* causing most pronounced conformity defects with improved cell wall physiology.

Effect of *C. albicans* infection on host immune system

The invasion of *C. albicans*, initially, activates the neutrophils, forming a part of the host immune system. These cells initiate variety of responses, such as, phagocytosis, degranulation, generation of reactive oxygen species (ROS), as well as release of the neutrophil extracellular trap (NET), to exert their defensive action. The formation of *C. albicans* biofilm, first, attracts the neutrophils to the site of infection, which are further stimulated to produce NETs. These are structures of DNA, histones, and antimicrobial proteins, well-suited for fighting large organisms, where phagocytosis alone cannot swallow large hyphal extensions (Kernien et al. 2018).

Biofilm associated with *C. albicans* infections are considered to be resistant to the attack by the monocyte, with an ability to change their cytokine profile. Peripheral Blood Mononuclear cells (PBMCs), in comparison to planktonic cells forbids the phagocytosis of biofilm associated with *C. albicans*. The down-regulation of TNF-, a cytokine that boosts phagocyte activation, is one of the most intriguing transitions, in this regard. Moreover, biofilm formed as a consequence of *C. albicans* infection, greatly influence the migratory ability of the macrophages, for, instance, literature reveals a reduction in the migration of murine macrophages by approximately twofold on exposure to the pre-formed biofilm (Alonso et al. 2017). Consequently, varied inhibitory responses corresponding to neutrophils and macrophages, associated with *Candida* biofilms may be elicited.

Anti-*Candida* drugs and their mechanism of action

The promising drugs available for the treatment of invasive fungal infections are meagre, in comparison, to those available for the treatment of diseases caused due to bacterial infections. The molecules contemporarily in clinical practice for the treatment of *Candida* infections are summarized in Table 1 with their mechanism of action corresponding to each class of drug and the existing cause of resistance.

A rigorous apprehension of the mechanism of action of various antifungal agents is imperative to further understand the resistance mechanism associated with them and is thereby, represented in Fig. 1. Presently, three major groups of antifungal drugs are typically used in clinical practices namely, azoles, polyenes and allylamines (or thiocarbamates), to fight the infections caused due to *C. albicans*. Almost all of these antifungal agents act by hampering the synthesis or interaction with a crucial component of the fungal cell membrane i.e., ergosterol (Prasad et al. 2016). This component is responsible for regulating the fluidity and integrity of the fungal cell membrane (Rodrigues 2018). Reported literature suggests the azoles primarily target the heme protein, involved in the co-catalysis of demethylation of lanosterol, mediated by cytochrome P₄₅₀ (Sobel and Sobel 2018). The inhibited C-14 α demethylation of lanosterol interferes with the ergosterol formation in the fungal cell (Zavrel et al. 2017). The affinities of azoles vary to their targets, accounting for the differences seen in their spectrum of therapeutic activity. Additionally, derivatization of the structure of azoles further contributes to the cross-resistance patterns observed in different *Candida* spp. (Yang et al. 2021). Unlike azoles, polyenes disrupt the fungal cells by inserting themselves into the fungal membrane through ergosterol binding, leading to the subsequent formation of porin channels. These aqueous pores ultimately cause the

outflow of intracellular ions, resulting in complete loss of transmembrane potential, impairing the cellular functions (Ortiz and Sansinenea 2019). Allylamines is another class of fungicidal drugs promoting the depletion of ergosterol content in the cell membrane through competitive inhibition of squalene epoxidase. This further occludes the conversion of squalene to lanosterol, leading to an increased squalene while a decreased ergosterol content in the fungal membrane (Hasim and Coleman 2019). Apart from these, several compounds have been delineated with the capability to directly affect the fungal cell walls (Chudzik et al. 2019). Aculeacins, echinocandins and paulacandins are the major cell wall inhibitors and best suited candidates for the treatment of *Candida* infections. However, among these, only echinocandins are clinically exploited for their in vivo and in vitro activities against species of *Candida* and *Aspergillus*, respectively (Su et al. 2023). The yeast cell wall is considered to be a multi-layered structure constituting of chitin, β -glucan and mannoproteins. Echinocandins exert their action through non-competitive inhibition of β -(1,3)-glucan synthetase, preventing the formation of glucan, a structural factor of the fungal cell wall, without exerting any prominent effect on the synthesis of nucleic acid or mannan component (Szymański et al. 2022). Glucan inhibitors secondarily affect other components of an intact cell by reducing the content of ergosterol and lanosterol and increasing the amount of chitin in the cell wall (Noel et al. 2023). This causes several cytological and morphological changes in the fungal cell such as growth of pseudo-hyphae, cell wall thickening, budding

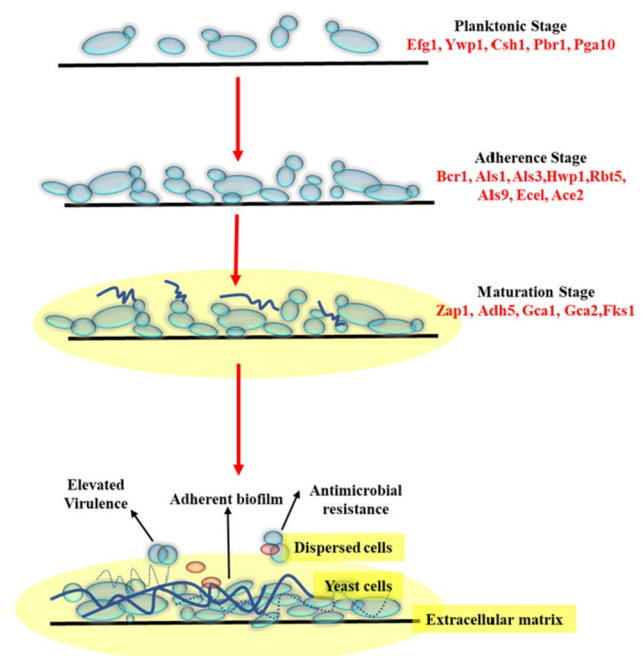


Fig. 2 Stages involved in *C. albicans* biofilm formation

(wherein buds are unable to detach from mother cells), and increase in osmotic sensitivity of the cells resulting into disintegration of the cell limited to the growing tips of the buds (Mathé and Dijck 2013).

Genome plasticity: a key to drug resistance in *C. albicans*

The genome of *C. albicans* is diploid and heterozygous, making it highly adaptive and compliant, with an ability to exhibit alterations in chromosomal rearrangement and aneuploidy. Genome plasticity plays a pivotal role in *C. albicans* adaptation and flexibility, thereby, possessing implications in *C. albicans* associated drug resistance, virulence as well as pathogenesis. The factors listed below are paramount genetic causes of arousing genome plasticity, promoting drug resistance in diseases related to *C. albicans* invasion.

- i. **Loss of heterozygosity (LoH) and aneuploidy:** The cells of *C. albicans* being naturally diploid, have eight heterozygous chromosomes, maintaining constant heterozygous nature. LoH comes into play due to the loss of information from one of the chromosomal homologues and hence corresponds to variability in virulence as well as fitness. LoH can copy the mutation of the second allele if a gene allele is mutated. LoH has been found in *CaTAC1*, *CaERG11*, and *CaMRR1* in clinical isolates of *C. albicans*, and it has been linked to increased resistance. Fluconazole resistance or the phenotypic changes in the fungal cells, including the appearance of grey colonies with reduced chances of phagocytosis by the macrophages and neutrophils, making them an ideal fit for colonizing the host, could all be reflected through the occurrence of LoH. Moreover, the rate at which it takes place is variable and is influenced by external factors (anti-fungal arsenals or high temperature). Furthermore, the defects arising during mitosis are a major cause of aneuploidy (an increase or decrease in the copy number of chromosomes). A key feature of aneuploid state is its potential of offering transient fix, during sex, promoting adaptation and hence the resistance. Various other factors such as, parasexual cycle as well as environmental stressors can also give rise to aneuploidy (Bhattacharya et al. 2020).
- ii. **Trisomy in chromosomes:** Usually, the gain of extra chromosomes in diploid cells is more common than the loss of the same, leading to trisomy disorders. One of the salient consequences of trisomy in chromosomes is the deregulation of cellular proteome that takes place in two levels. Initially, the genes carrying the additional chromosome are expressed, resulting

in an elevated mRNA as well as protein abundance. Hence, the presence of an excessive individual chromosome results in only a modest elevation in the level of proteins, indicating a mere phenotypic effect due to over-expression of individual genes as a consequence of trisomy. In the therapeutic isolates, neither efflux pump behaviors nor altered ergosterol biosynthesis could be blamed. Resistance development due to trisomy in chromosome 4 still needs to be worked upon and is directly linked to increase in azole tolerance. Trisomy on chromosome 3 and chromosome R causes improved triazole tolerance and develops on prolonged exposure to azoles. However, prolonged exposure of fluconazole corresponds to trisomy of chromosome 6, with increased expression of *Mdr1* gene, over-expression of which during mid-logarithmic phase of growth plays a minor role in fluconazole resistance. Additionally, trisomy in chromosome 2 ($\text{Chr2} \times 3$) and chromosome 5 ($\text{Chr5} \times 1$) causes caspofungin resistance, the mechanism of which is yet to be known (Bhattacharya et al. 2020).

Overcoming resistance as an adjunct to drug discovery

The emergence of anti-fungal resistance in *Candida* species is predominantly promoted by the factors listed below and hence surpassing these barricades would eventually open avenues for the discovery of novel drugs.

Biofilm formation

Researchers have elucidated biofilm formation to be one of the paramount causes of anti-microbial Resistance in *C. albicans*. Cells of the pathogenic fungi growing in the biofilm are known to be more resistant to the commercially available fluconazole by a factor of approximately 100 folds, while the same cells show an increased resistance to amphotericin B by a factor of 20–30 fold in contrast to when these cells are present in their planktonic form. *C. albicans* biofilm formation involves several phases, including, the adhesion of yeast cells to the substrate, proliferation of these cells to form fungal culture, synthesis of extracellular matrix and hyphae formation fostering the attachment of the proliferated cells. The stages involved in the formation of *C. albicans* biofilm are illustrated in Fig. 2. The extracellular matrix, enveloping the fungal cells is the foremost thing to encounter the immune cells. The biofilm matrix is composed of proteins (around 55%), one of the most abundant macromolecule forming a part of the biofilm matrix, followed by carbohydrates, lipids and other genetic material. A large number of the matrix

Table 2 Proteins and genes involved in *C. albicans* biofilm formation

Protein Name	Gene reference (Orf19#)	Function of the desired protein	References
Positive regulation both under in vitro and in vivo conditions			
Ace2	Orf19.6124	Transcription factor Governs development of biofilm	Granger 2012)
Bcr1	Orf19.723	Transcription factor Governs biofilm formation by promoting cellular adhesion	Morschhäuser 2002)
Cas5	Orf19.4670	Transcription factor Governs biofilm formation by promoting cellular adhesion	Granger 2012)
Cdr1 and Cdr2	Orf19.6000 and Orf19.13379	Multi-drug transporter Governs biofilm mediated drug resistance	Singleton et al. 2005)
Csh1	Orf19.4477	Cell–matrix adhesion regulator Negatively regulates the production of matrix with the activation of ZAP1	Langford et al. 2013)
Czf1	Orf19.3127	Transcription factor Governs adherence of yeast cells	Li and Palecek 2003)
Eap1	Orf19.1401	Cell wall protein Governs cell to cell adhesion of <i>C. albicans</i>	Cao et al. 2006)
Efg1	Orf19.610	Transcription factor Governs cell adhesion and hyphae formation	Li and Palecek 2003)
Flo8	Orf19.1093	Transcription factor Governs hyphae formation	Du et al. 2012)
Gat2	Orf19.4056	Transcription factor Governs biofilm formation	Nobile et al. 2006)
Hwp1	Orf19.1321	Cell-wall protein Governs yeast cell adhesion and interaction of epithelial cells	Tsai et al. 2014)
Mss11	Orf19.6309	Transcription factor Governs the growth of hyphae in <i>C.albicans</i> biofilm	Tsai et al. 2014)
Mkc1	Orf19.7523	MAP kinase Governs biofilm formation, structure of the cell wall as well as filamentation in yeast cells	Branco et al. 2021)
Ndt80	Orf19.2119	Transcription factor Assists in biofilm formation and hyphae development	Braun et al. 2001)
Nrg1	Orf19.7150	Negatively governs the filamentation of the yeast cells	Delgado-Silva et al. 2014)
Rlm1	Orf19.4662	Positively governs the development of extracellular matrix	Glazier et al. 2017)
Rob1	Orf19.4998	Transcription factor Governs biofilm formation	Banerjee et al. 2008)
Tec1	Orf19.5908	Transcription factor Governs the formation of hyphae and biofilm in <i>C.albicans</i>	Panariello et al. 2017)
Ume6	Orf19.1822	Transcription regulator Governs filamentation of yeast cells	Kim et al. 2019)
War1	Orf19.1035	Transcription factor Governs the adherence of yeast cells	Mukherjee et al. 2006)
Arg81	Orf19.4766	Transcription factor Auxiliary biofilm transcription factor involved in adhesion and filamentation stage	Finkel et al. 2012)
Efg1	Orf19.610	Transcription factor Core transcription factor, governing the adhesion and filamentation stages of biofilm formation	Glazier 2022)
Negative regulation both under in vitro and in vivo conditions			
Adh1	Orf19.3997	Alcohol dehydrogenase Limits development of biofilm via an ethanol-dependent pathway	Hronek et al. 2005)
Ywp1	Orf19.3618	Yeast cell wall protein Might play a role in host dispersal and its mutation would lead to an elevated yeast cell adhesion	Morschhäuser 2002)
Zap1	Orf19.3794	Negatively regulates cell matrix development	Tsai et al. 2014)

Table 2 (continued)

Protein Name	Gene reference (Orf19#)	Function of the desired protein	References
Spt7	Orf19.7572	SAGA complex subunit regulating the filamentation and invasiveness in <i>C. albicans</i>	Rashid et al. 2022)

components are remarkably different from the components of the cell wall of normal *C. albicans* cells and thus entice the host immune system of desired response. Additionally, the composition of this matrix differs to a great extent, when analyzed in vivo models. Thus, the analysis of *C. albicans* biofilm through animal model is a pressing priority.

The genes and their corresponding proteins implicated in the biofilm formation associated with *C. albicans* infection are elaborated in Table 2.

Development of vesicular vacuoles

The development of vesicular vacuoles assists to get rid of the foreign material by sequestering the toxic compounds. These have a vital role to play in fluconazole resistance, exhibited by *C. albicans*.

Persister cells

These refer to the genetically similar sub-population of the fungal pathogens that are fleetingly resistant to antibiotics. In other words, these cells are incapable of growing in the presence of antibiotics, however, have the potential to withstand the attack by these entities up till they reside in the persister state. Despite of being dormant in nature, these cells can resume growth on removal of the antibiotic. There exists various mechanisms inducing dormancy in such cells, assisting them to survive the antibiotic action. Researchers have profusely illustrated an informal association of protein aggregation with different forms of fungal dormancy, specifically persistence in various pathogenic cells, including *C. albicans*.

Efflux pumps

The membrane proteins liable for the transportation of substances across the cell membrane, constitute of the efflux pumps. The fungal efflux pumps are known to transfer xenobiotics including certain fluorescent compounds, enabling the utilization of fluorescence-based detection along with the assays inhibiting growth, with subject to discovering the compounds targeting anti-fungal drug resistance through efflux-mediated route. The efflux proteins, forming a part

of the ATP binding cassette (ABC) and major facilitators (MFS) super family, are prime contributors of the multi-drug resistance, prevalent in yeasts. Efflux pump-mediated resistance generally hampers the action of azoles for the treatment of *Candida*-related infections.

Alternatives in treatment of *C. albicans*-related infections

The evolution of anti-*Candida* drug resistance is a primary obstacle in the treatment of these infections in clinical settings. Hence, the progression of novel or alternative anti-fungal molecules, as well as, transformation of the prevailing drug molecules is vital for the effective treatment of the *C. albicans*-related infections, in a less toxic manner. The alternatives can be extracted from different sources including plants, animals, or even other fungi. Herein, we attempt to abridge the ill-effects of the existent antifungal therapies, and summarize the discoveries associated with the development of novel promising avenues for curing fungal infections.

Regardless of the sustained and cumulative progress in the field of anti-fungal drug development, there exists impending threats concomitant with the use of prevailing drugs and the development of the novel compounds, with a boost in the cases relating to AMR, owing to the over-prescription of drugs and the property of biofilm formation exhibited by these lethal pathogens. The intricacies, such as, neurotoxicity, biochemical abnormalities (azotemia), electrolyte imbalance, liver injuries, and compromised renal concentrating ability, associated with the use of azoles amphotericin B, lead to an upsurged demand for novel alternatives in the treatment of *C. albicans* infections. Additionally, limited bioavailability, along with restricted pharmacokinetics of the commercially available drugs, elevates the stipulation for the establishment of alternative treatments of lethal fungal infections.

Table 3 complies the alternatives to the customary routes of treating infections related to the invasion of various *Candida* spp., with a special focus on *C. albicans*.

Moreover, the establishment of novel agents, specifically targeting fungal cells, is a primary requirement for the development of effective anti-fungal therapies. So far, fungal inhibitors such as olorofim (act by impairing fungal pyrimidine biosynthesis pathway), VT-1161 (active against *C. albicans*), T-2307 (broad spectrum anti-fungal agent), etc. are under several stages of clinical trials and potential anti-fungal arsenals for near-future (Wiederhold 2018). Synergistic combination

Table 3 A precise account of the alternative anti-*Candida* treatment available

Alternative treatment	Benefit over the existing therapies	References
Diet	The growth of <i>C. albicans</i> is promoted by refined sugars (sucrose, fruit juice, and honey). Involving more of garlic, that has fungicidal effect, nuts with required fatty acids, and cloves, an antifungal agent, tends to get rid of the infection. Thus, <i>Candida</i> diet (a diet without alcohol, simple sugars, yeast as well as only a restricted amount of processed foods) is healthful and a natural way of preventing such infections. Involving probiotics in the diet prevents the growth of this pathogenic microbe	Martins et al. 2014)
Immune-strengthening remedies	The intake of vitamins, including Vit. C, Vit. E and Selenium curtails inflammation, making the immune-system and strong and sound. Stress and dietary controls also contribute their bit in strengthening of the immune responses. Furthermore, consumption of sufficient amounts of antioxidants (carotene, Vit. A, and zinc) assists in accelerating the functions of thymus gland, increasing the production of T-cells that help our immune system to get rid of infectious diseases	Reddy and Nancharaiah 2020)
Ionic Liquids (ILs)	ILs are promising synthetic assets in eradicating the <i>C. albicans</i> pre-formed biofilms preceding the treatment and thus are seen as boon for fighting fungal infections. However, only a limited apprehension of the mechanism of action associated with the use of ILs as antifungal agents, limits their applications in drug discovery. The evidences in literature regarding the employment of imidazole-based cationic moieties in curbing the fungal biofilms have divulged the attention of the researchers towards these green-alternatives. They can prove to be promising bio-compatible solvents, exhibiting great synergism with the prevailing commercially available drugs	de Andrade Monteiro, Cristina, J. Ribeiro Alves dos Santos 2019)
Phytochemicals	Certain plant-derived natural compounds (phytochemicals) portray anti-fungal activities at lower concentration, within permissible limits without manifesting any toxic effects over the host organism. These can be potential drug molecules in fighting microbial infections	Dwivedi et al. 2020; Sharma et al. 2023)
Nanoparticles	Nanoparticles of silver, gold, silica and selenium have proven to possess anti-fungal properties. It is postulated that these nanoparticles act by rupturing the fungal cell wall and cell membrane and actively targets their ergosterol content and fatty acid composition. The nanoparticles of chitosan (a biopolymer) are also effective against <i>C. albicans</i> and various other fungal strains. Many nanoparticles (carbon-based nanomaterials, hydroxy apatite, ZrO ₂ & CuO silica, TiO ₂ , Ag ₂ O, ZnO) have been appreciated potential anti- <i>Candida</i> agents. Nanoparticles have a better fungicidal effect than conventional anti-fungal drugs, owing to better penetration in host cells even in little concentrations	Ahmad et al. 2020)
Photodynamic therapies	The drug-free treatment of <i>Candida</i> biofilms can be brought about by the employment of photo-sensitizing compounds. It has been reported that photosensitizers such as methylene blue and toluidine blue can destroy <i>C. albicans</i> biofilm, imparting great synergism with chitosan. These agents are nontoxic as well as economical, as compared to the conventional antifungal medication	Hamblin and Abrahamse 2019)

Table 3 (continued)

Alternative treatment	Benefit over the existing therapies	References
Fungal vaccines	Contemporarily, there have been no licensed vaccines, within reach, for controlling and combating fungal infections, owing to high manufacturing cost, toxicity issues, and expensive clinical trials. However, in animal models, the progress has been remarkable, since various vaccine designs (including sub-unit formulations or even live attenuated fungus), have been employed to achieve immunity against the primary, clinically relevant mycoses. The concept of pan-fungal vaccines is a holy grail, targeting fungal cell wall, that is speculated to provide protection against a broad-range of systemic fungal morbidities commonly occurring in the human population	Oliveira et al. 2021; Nami et al. 2019)

therapies, instead of single drug-based mono-therapies with ILs as well as other bio-compatible molecules have recently attracted the attention of researchers worldwide (Wu et al. 2020).

Conclusion

The increment in the cases associated *Candida* infections is petrifying, and a cause of morbidity and mortality, across the globe. *Candia spp.*, being extremely resistant to antibiotics, have tendencies to acclimatize to a variety of host niches, jeopardizing human health to a large extent. In this review, we made an attempt to summarize the causes and consequences of *Candida*-related infections with a special focus on *C. albicans* as a model organism. Recent breakthroughs in the diagnosis of different *Candida* species have been discussed and their benefits over the customary diagnostic techniques have been highlighted in the present work. This review also points up, biofilm formation in *Candida spp.* as a fundamental component of the disease pathogenesis. The distinctive features, such as, anti-fungal drug resistance, ability to provide defiance against the components released from the host immune system, stress-tolerance and their ability to promote the spread of new infections, associated with the formation of *Candida* biofilms is of primary concern in clinical settings. Therefore, the review emphasis on the need to explore the voyage of science for the discovery of novel agents, capable of fighting anti-fungal infections. Various alternatives have been provided in the last section of the review. These alternatives alone or in combination with the conventional therapies can prove to be a cornerstone in the treatment of yeast infections.

Acknowledgements Authors gratefully acknowledge the financial support of grant received from ICMR, Government of India, Delhi-110029 [No./ICMR/52/06/2022-BIO/BMS]. Kajal Sharma acknowledges the financial support received from Department of Science and Technology through INSPIRE scheme (IF200397).

Author contributions K.S. wrote the main manuscript and prepared all the figures. P.K.P. wrote the main manuscript and M.S. reviewed and supervised the manuscript writing. All the authors reviewed the manuscript.

Funding The authors have not disclosed any funding.

Declarations

Conflicts of interest There is no conflict to declare.

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